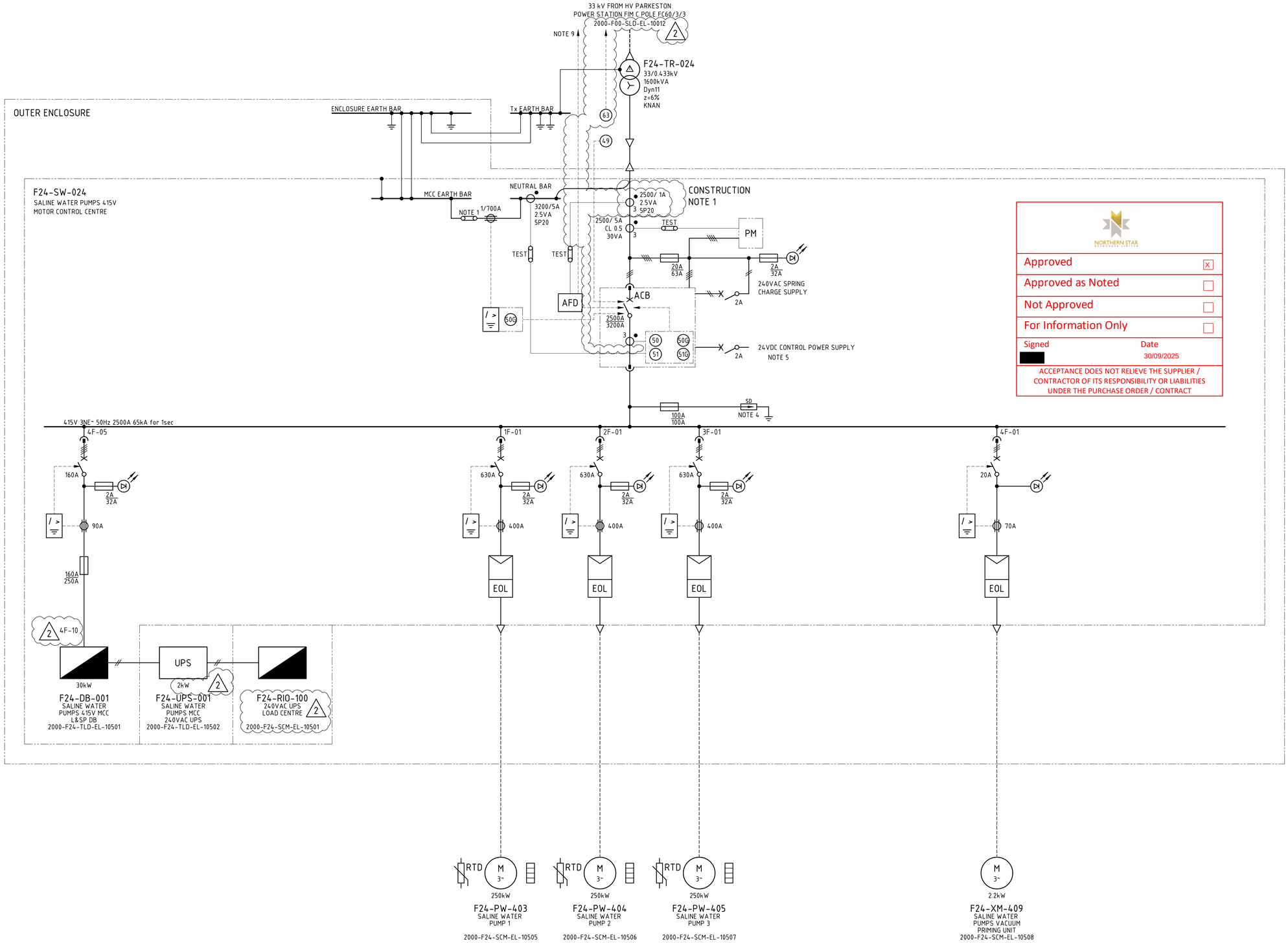


NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 30/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F24-SW-024 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
3. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F24-BLK-IC-10501.
4. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
5. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
6. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
7. ANY DRIVE RTDs SHALL BE WIRED TO REMOTE I/O CARDS. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
8. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
9. (2) UPSTREAM TRANSFORMER FEEDER (RECLOSER) SHALL BE TRIPPED ON DETECTION OF TRANSFORMER PRESSURE RELIEF OR ARC FLASH DETECTION.

CONSTRUCTION NOTES

1. ARC FLASH PROTECTION CT'S TO BE INSTALLED ON SITE.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 2, 3
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 2, 3
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 2, 3
- THERMAL OVERLOAD UNIT NOTE 6
- VARIABLE SPEED DRIVE NOTE 2, 3
- EARTH LEAKAGE RELAY NOTE 2, 3
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- (49) (63) (71) — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 8)
- xxxA ST TRIP SET POINT; xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- PM — POWER METER

DRG No.	REFERENCES
2000-F24-SCM-EL-10503	F24-SW-024 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F24-SCM-EL-10504	F24-SW-024 ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F24-SCM-EL-10501	F24-SW-024 INCOMER SCHEMATIC DIAGRAM SHEET 1
2000-F24-SCM-EL-10502	F24-SW-024 INCOMER SCHEMATIC DIAGRAM SHEET 2
2000-F24-DRG-EL-10501	F24-SW-024 ELECTRICAL EQUIPMENT LAYOUT
2000-F24-BLK-IC-10501	F24-SW-024 AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F00-SLD-EL-10011	OVERALL SITEWIDE 33kV DISTRIBUTION SINGLE LINE DIAGRAM SHEET 2
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	09SEP25	PROTECTION SYSTEM UPDATE					
1	22JUL25	33kV OHL POLE NUMBER REVISED, FAT REDLINES INCORPORATED					
0	12NOV24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

PRIMERO
primero.com.au | DESIGN | CONSTRUCT | OPERATE

PRIMERO PROJECT NUMBER: 23517

INFORMATION AND REPRESENTATIONS ON THIS DRAWING ARE THE INTELLECTUAL PROPERTY OF PRIMERO UNLESS SPECIFICALLY IDENTIFIED OTHERWISE. ALL RIGHTS INCLUDING USE RESERVED

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: AIR & WATER SERVICES - SALINE WATER
F24-SW-024 415V MCC
SINGLE LINE DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F24-SLD-EL-10501	2

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELEC\F24 - AIR & WATER SERVICES\2000-F24-SLD-EL-10501.DWG 16SEP25 4:05PM HOTSEAT04